worked to enrich the others in the team and experience the enjoyment of working together.

The primary objectives of the SLV Project were design, development and operation of a standard SLV system, SLV-3, capable of reliably and expeditiously fulfil ing the specified mission of launching a 40 kg satel ite into a 400 km circular orbit around the earth.

As a first step, I translated the primary project objectives into some major tasks. One such task was the development of a rocket motor system for the four stages of the vehicle. The critical problems in the completion of this task were: making an 8.6 tonne propel ant grain and a high mass ratio apogee rocket motor system which would use highenergy propel ants. Another task was vehicle control and guidance. Three types of control systems were involved in this task—aerodynamic surface control, thrust vector control and reaction control for the first, second and third stages and the spin-up mechanism for the fourth stage. Inertial reference for control systems and guidance through inertial measurement was also imperative. Yet another major task was the augmentation of launch facilities at SHAR with systems